

Omdia Universe: AI-assisted Software Development, Part 1: IDE-based Tools, 2026

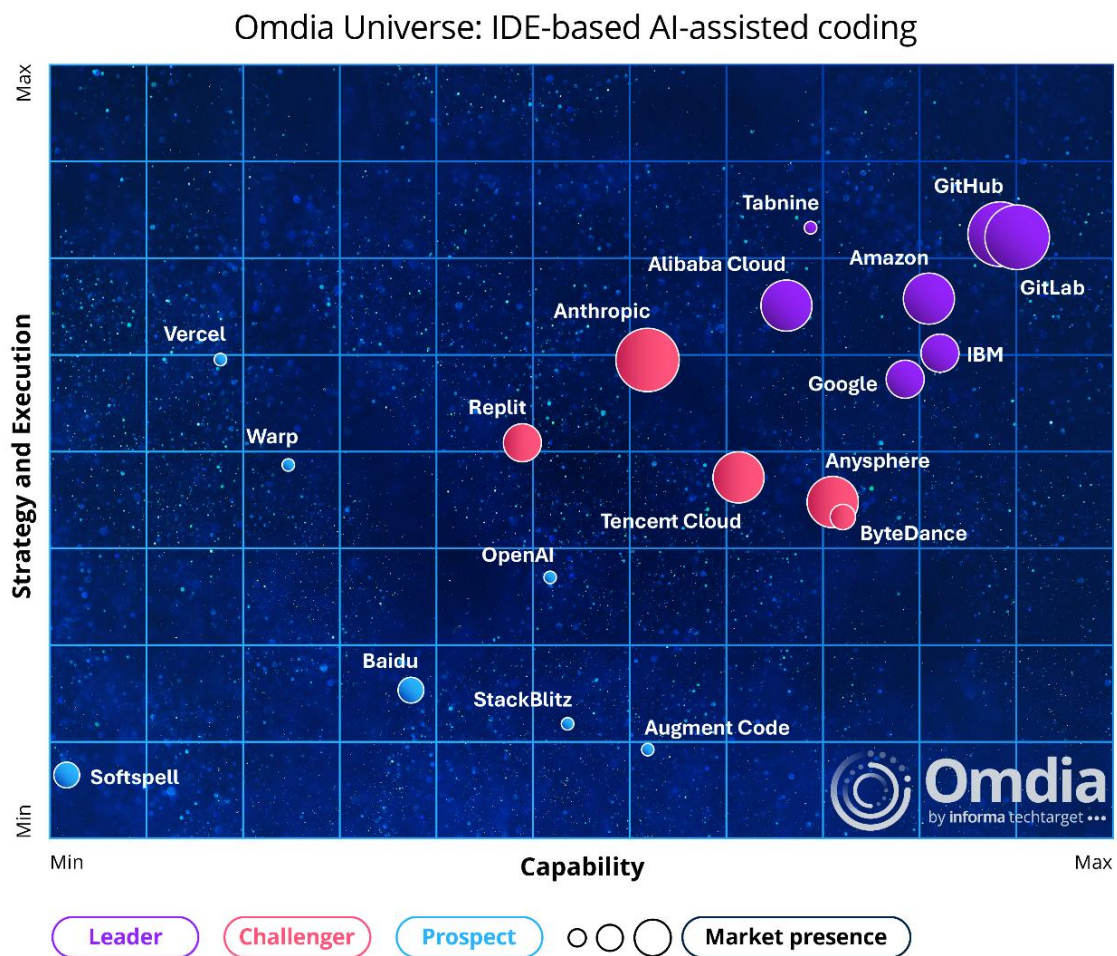
IBM

Summary

Catalyst

AI-assisted tools have completely transformed software development, as the reliability of code generated by frontier large language models (LLMs) has reached a level of accuracy that allows application developers to let the machine generate the bulk of application code. The implications of this technology shift are examined in this Omdia Universe, with market segmentation analysis, and an assessment of the leading tools that plugin to integrated development environments (IDEs).

Figure 1: The Omdia Universe for AI-assisted Software Development, IDE-based Tools, 2026



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Source: Omdia

All vendors included in the report were approached to provide input for this study, of which 50% participated. Anthropic, AnySphere, Augment Code, Baidu, Google, OpenAI, Replit, StackBlitz, and Warp declined to participate or did not complete the response questionnaire in time before publication. The non-participant vendor evaluations are based on publicly available information,

analyst insights, and feedback from their customer base sourced directly via Informa TechTarget audiences.

Omdia view

Introduction

The field of AI-assisted software development is evolving rapidly, which is why Omdia is now covering this topic annually in its Universe. The market can be divided into essentially three segments:

- **IDE-based tools:** The AI assistant tool either comes with its own IDE and/or plugs into popular IDEs such as Visual Studio (VS) Code and IntelliJ IDEA. While the IDE is the natural home of the professional developer, and these tools are generally targeted at this role, it is possible to create applications in a “no code” mode in which the user provides application requirements and the machine generates the code. Code review, testing, and deployment can be conveniently performed in the IDE.
- **Pure no-code tools:** This category of tools is targeted at users with business, engineering, or management roles and requires no programming experience. The user inputs the requirements, and the machine automates all aspects of application generation.
- **No-low-pro AppDev platforms:** Where the IDE has a typical environment geared towards professional developer needs, in the AppDev platform category, the provider has the freedom to create its own unique environment, typically emphasizing ease of use. AppDev platforms typically span no-code and low-code application development, especially when a graphical environment is used to define the application. They also target professional developers skilled in the graphical method. Hence, Omdia labels this tool as spanning “no-low-pro” skill sets.

This report focuses solely on IDE-based tools, while a forthcoming companion Universe focuses on AppDev platforms.

AI assistant tools

An AI assistant tool comprises two essential elements: the LLM model and the harness that surrounds the LLM and provides the interface between the user and the LLM. An important function of the harness is to enhance the context input to the LLM to provide the best possible generated output. Examples of context enhancement include providing retrieval-augmented generation (RAG) to enhance the LLM with a private codebase, providing context prompts on organizational coding standards and architectural compliance, and tapping into organizational knowledge bases to provide relevant information. One vendor in this report, Tabnine, has made its Context Engine a separately available product to work with competing tools.

The vendors in this report vary in the LLMs they offer to the tool user; some work with only a select number of frontier labs. The latest releases of top LLMs from the key labs narrow the difference between them, but most importantly, have raised the level of reliability and accuracy to the extent that developers interviewed for this report talk of a phase change, where the role of the human developer is more to supervise the AI assistant as it generates the bulk of code. At the time of writing, the three top LLMs for coding that are consistently mentioned are:

- **Anthropic:** Claude Opus 4.6—a lighter version, Sonnet 4.6, is also recommended for its lower cost and suitability for long-running tasks. Claude Opus 4.5 was described as a “game changer”; rival LLMs and v4.6 from Anthropic swiftly followed.
- **OpenAI:** GPT 5.2 Codex—the Codex variant is heavily optimized for coding agents, large refactors, migrations, and terminal-based tasks.
- **Google:** Gemini 3.1 Pro is praised for natural interaction, understanding entire repositories, and strong agentic performance.

Note that this report does not assess the LLMs but rather the AI assistant tools built on top of the frontier models.

The changing role of the human software developer

While the current generation of AI-based development tools cannot wholly replace software developers, they do improve their productivity—applications that took weeks or months to develop can now be created in a fraction of the time. Another aspect is how AI can reduce delays. The software development lifecycle (SDLC) workflow can contain bottlenecks; for example, on completing a task, the developer needs to find the right skilled developer to review it, thus creating a delay before the code can be pushed into the mainline. AI tools can automate code reviews and other tasks that delay the workflow.

While LLMs can be prone to hallucinations, these are beginning to be reduced, and one approach is to use a different LLM to perform code reviews. With AI tools handling the bulk of coding, the human’s role is to ensure requirements are met, that any issues or bugs are addressed, and to provide essential guidance. Many organizations will likely exploit the AI-led rise in productivity by reducing the developer headcount. However, there are essential roles that machines cannot replace, for example, supervising the coding agents and then driving QA, testing, and debugging, so headcount may only drop marginally.

The changing role of the virtual AI agent

While the task of coding is specific within the SDLC, there are many handoffs to specialist skills across the SDLC and even within coding itself: backend networking, data services, middleware, frontend design and development, etc. While AI assistant tools take on the role of a programmer, i.e., a virtual software engineer, the latest generation of agentic AI tools has a virtual software engineering *team* aspect. An orchestrating AI agent farms out tasks to specialist virtual agents and coordinates the completion of a major task across multiple agents. While this technology is still in the early stages of development, some vendors already offer these features. The trend is for organizations to have development teams consisting of virtual and human engineers, with the humans supervising and driving the virtual workers.

Analyzing the AI-assisted software development: IDE-based tools universe

Market definition

While Omdia has repeated this report from last year, it has updated the assessment criteria for solution capability. Meanwhile, Strategy & Execution (the y-axis of the Universe chart) remains largely unchanged. However, the section on technology innovation has been updated with questions about agentic AI capabilities.

Solution capability comprises the tool's technical features. This year, Omdia assessed applicability across the entire SDLC, from requirements management tasks, planning, and design to team and issue management. As the tools take on agentic AI capabilities, they can perform tasks across the SDLC and not just coding, hence the broader scope of the assessment.

Here, Omdia delves deeper into the solution capability assessment. Refer to the appendix for Omdia's definition of Strategy & Execution.

Core features

- **Business requirements** - Can the tool generate formal, structured requirements from unstructured documents, filter the requirements for ambiguity, duplication, and more?
- **Planning/design** - Generate user stories from requirements, user story estimation, and more.
- **Specifications** - Can the tool ingest unstructured or semi-structured business requirements (e.g., Word docs, PDFs, Jira tickets, meeting transcripts), and more?
- **Coding** - Will produce large-scale code to complete applications from natural language (NL) prompts, and more.
- **Testing** - Generate tests from requirements with full traceability, and more.
- **Code reviewing** - Types of code reviews supported: pull request reviews, security audits, performance optimization, refactoring suggestions, style/standards compliance, and more.
- **Documentation** - Types of documentation an AI tool can generate: inline code comments, API reference documentation, architecture diagrams, user manuals, README files, and more.
- **Knowledge base** - Types of knowledge sources that can be ingested: internal wikis, collaboration/chat tool archives, issue/bug tickets, code comments, meeting transcripts, design docs, and more.
- **Deployment/DevOps** - Supports CI/CD pipeline automation, automated incident resolution, and more.
- **Team management** - Managing agile projects, integrating bidirectionally with project management and ticketing tools, and more.

- **Issue management** - Issue ticketing system, automated triaging, prioritizing issues, assigning to team members, a dashboard with summarization and other metrics, and more.

Extended features

- **APIs** - Creating API and data services: API management and gateway features, auto-generation of SQL based on plain text, SOAP and REST services/API, web services, GraphQL, gRPC, MCP server, and more.
- **Agile Dev** - Advanced agile practices: auto story splitter for large stories, auto “Definition of Done” checklist enforcer, smart acceptance criteria generator from raw user story, live cycle-time predictor, one-click refactor and test, pull-request description and auto changelog, live AI pair programmer, can drive keyboard when commanded, and more.
- **Security** - Supports developing security and access control in applications: HTTPS, online authentication, offline authentication, system for cross-domain identity management (SCIM), and more.
- **AI/machine learning (ML)** - Has measures to minimize hallucinations, has guardrails to control AI output, generated code free of the Open Worldwide Application Security Project (OWASP) top 10 vulnerabilities, has an AI/ML development studio, supports AI/ML feature engineering, offers prebuilt AI services, and more.

Market dynamics

The AI-assisted tool market, IDE-based category, consists of plug-ins to popular IDEs or provided IDEs (typically forks of the open source VS Code). As discussed above, these tools provide essential harnesses that perform context engineering, feeding crafted prompts to the LLM. The way the tool internally interacts with the LLM is also superior to, say, a developer text chatting/prompting directly with a frontier LLM. The harnesses exploit native function calls (NFCs) available through the LLM API. An LLM may need to call on other tools and libraries to complete a task. The NFC provides a robust calling protocol, generating structured outputs (typically JSON objects) that specify which external function/tool to invoke and the exact parameters, rather than relying solely on free-form text generation or heavy prompt engineering to simulate this behavior.

NFCs deliver much higher reliability and consistency in producing valid, correctly formatted structured data (often 80–95%+ success rates on leaderboards versus 50–70% with prompting alone, especially on complex or multi-tool scenarios), reduce hallucinations or malformed outputs that require extra parsing/error-handling logic, enable parallel/multiple tool calls more naturally in a single response, lower token usage and latency by avoiding verbose explanatory text, and make integration into applications far simpler and more robust since the output is directly executable without brittle regex/JSON repair steps. NFCs transform LLMs from pure text generators into dependable agents that can more accurately bridge NL to programmatic actions.

Omdia recommends developers not use code generation directly with LLMs for enterprise-grade applications because of the benefits of NFCs when working with agentic workflows and for improved security vetting of generated code.

Omdia reached out to the top 20 IDE-based AI-assisted development tools. The only popular tool not to appear in the Universe is Windsurf, which Cognition, the maker of Devin AI, acquired. The Cognition

portfolio of tools appeared too low in the Universe chart and was removed from the lineup, leaving 19 vendors assessed in this report.

Figure 2: Vendor rankings in the AI-assisted Software Development: IDE-based Tools Universe

Vendor	Product(s) evaluated
Leader(s)	
Alibaba Cloud	Alibaba Cloud Qwen, <u>Lingman</u> , and Model Studio
Amazon	Kiro
GitHub	GitHub Copilot
GitLab	GitLab Duo Pro, GitLab Duo Enterprise, and GitLab Dio Agent Platform
Google	Gemini Code Assist
IBM	IBM Bob
Tabnine	Tabnine, Context Engine
Challenger(s)	
Anthropic	Claude Developer Platforms
<u>Anysphere</u>	Cursor
ByteDance	TRAE
<u>Replit</u>	<u>Replit Agent</u>
Tencent Cloud	Tencent Cloud <u>CodeBuddy</u>
Prospect(s)	
Augment Code	Agent
Baidu	Baidu Comate
OpenAI	Codex
<u>Softspell</u>	<u>Softspell</u>
<u>StackBlitz</u>	<u>Bolt.new Code</u>
<u>Vercel</u>	v0
Warp	Agents

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Source: Omdia

Market leaders

The market leaders have performed well on both axes of the Universe chart: Strategy & Execution and Solution Capability. All the Omdia leaders have market recognition for their enterprise-oriented solutions in this field. Two of the leaders—Anysphere and Tabnine—are pure play vendors, while the others offer AI-assisted development tools as part of their portfolio of solutions. On the technology features side, Omdia added features across the end-to-end SDLC, given the increasing use of agentic AI, the capabilities of these solutions are increasing in this respect, and the leaders are separated from the challengers by demonstrating these features. Omdia recommends that leaders should automatically appear on a prospective customer's shortlist.

Market challengers

Challengers can be more focused on particular market segments, with a slightly narrower feature range, or are earlier in their maturity cycle and have yet to build out the full spectrum of features that leaders possess. With the right market momentum and feature maturity, Omdia expects some challengers to move up to the leader ranking. In its opinion, challengers should be shortlisted when their strengths match the customer's requirements.

Market prospects

In general, there are different types of prospect vendors:

- They may offer legacy solutions that have not kept pace with the market.
- They may be startups that lack the breadth of features being assessed but have a roadmap and intention to follow the leaders and (if successful) enter the next ranking level.
- They may offer solutions that overlap with the solution space being assessed. In other words, this is not their sweet spot; rather, it is the nearest technology assessment space that covers their solution.

In this report, all the prospects have targeted the AI-assisted coding space well, but lack some of the breadth of support that is being assessed across the SDLC. Depending on their needs, customers may find suitable solutions in this ranking category and should drill deeper to match their requirements to the available tool features.

Non-participating and partially participating vendors

Some vendors cannot fully engage; nevertheless, given their market presence, Omdia aims to assess their offerings based on publicly available information and Omdia analyst insights. In this report, such vendors are Anthropic, Anysphere, Augment Code, Baidu, Google, Replit, StackBlitz, and Warp. Amazon was only available for fact-checking.

Market opportunities

The use of AI-assisted software development is now mainstream, with either dedicated AppDev platforms or IDE plug-ins; developers cannot miss the presence of these tools. Enterprises are adopting these tools to improve the productivity of their workforce, and the role of software developers is changing. The reliability of code generated by leading frontier LLMs has increased to the point that organizations have greater confidence in using it. Consequently, there will be increasing opportunities for businesses that can adapt their software development capability to work with the new generation tools, as AI offers a leap in productivity. Knowing how best to exploit AI and how best to use the human developers will be key. Omdia does not think that making software developers redundant is the right answer; it is better to employ them to supervise the AI tools and mediate between business stakeholders and the AI tools.

Market threats

A pitfall for organizations when adjusting to the new generation of AI tools is a misunderstanding of the role that still lies with software developers. There will be a learning phase on how best to apportion tasks between humans and machines. Omdia believes human developers will need to remain in the driving seat to make the best use of AI, especially to ensure business value and requirements are met and to test AI-generated code. For software developers, AI does pose a threat to job security. Omdia

recommends that software developers acclimate to the new generation of tools and learn to become more productive using them.

AI poses a threat to businesses that fail to embrace the opportunities that AI-based application development can offer. Consequently, they lose market share to faster-moving competitors. The AI genie is out of the box and cannot be put back in—it is now a question of learning how to exploit this technology to best advantage to stay competitive in the market.

Market outlook

Given the pace of evolution in frontier models and the AI tools that harness them, Omdia believes that the capability of AI-assisted development tools will continue to grow. The growth will encompass coverage of the end-to-end SDLC and agentic AI capabilities, whereby the AI tool can be instructed to build a complete application from requirements. Nevertheless, Omdia recommends not to forgo the lessons learned from agile development—software development is as much about human-to-human communication as it is about computer programming. Understanding business requirements and what delivers value to the business is crucial. Therefore, the role of human developers will remain necessary for the current and near-term generations of AI.

Vendor analysis

Note: Vendor profiles are only provided on participating vendors.

Vendor accolades

Within the vendor analysis section, two types of accolades can be awarded to vendors:

- The **Best in class** accolade is awarded to the vendor(s) with the highest score (highest outright, tied highest, or within <1% of the highest score) for each of the scoring categories that make up this Universe topic:
 - Core development
 - Extended features
 - Solution breadth
 - Strategy and innovation
 - Market momentum
 - Vendor execution
- The **Top-tier** accolade is given to vendors falling within the upper tercile (top third) of the scores within the comparison group for each of these same scoring categories.

IBM (Omdia recommendation: Leader)

IBM should appear on your shortlist for its advanced solution targeted at enterprise users.

Overview of IBM Bob

IBM Bob, an AI-first integrated development environment and modernization assistant, represents IBM's advanced entry into AI-assisted software development, particularly tailored for enterprise-scale challenges. Bob positions itself as more than a traditional code completion tool; it is designed as an AI development partner that understands developer intent, full repository context, organizational coding standards, and security and compliance requirements. It includes its own IDE based on VS Code, and Bob-Shell extends the same AI assistance to a terminal. Bob offers agentic workflows that enable AI to proactively plan tasks, generate to-do lists for approval, execute multi-step processes (e.g., design, code, refactor, test, document), perform intent-aware code reviews, detect vulnerabilities early, and accelerate modernization efforts. Internal IBM adoption has been strong, with more than 40,000 IBM developers reportedly achieving roughly 45% productivity gains by automating mundane tasks and focusing on higher-value work.

Key differentiators include the following:

- Deep enterprise readiness
- Automated selection of a foundation LLM based on tasks and cost-benefit analysis
- Using IBM's Granite foundation models that have training data non-infringement guarantees
- Fine-tuning for domain-specific tasks
- Guardrails to minimize hallucinations and ensure secure outputs
- Specialized strengths in enterprise application modernization, including COBOL, PL/I, RPG, and Assembler on mainframe IBM Z and IBM i

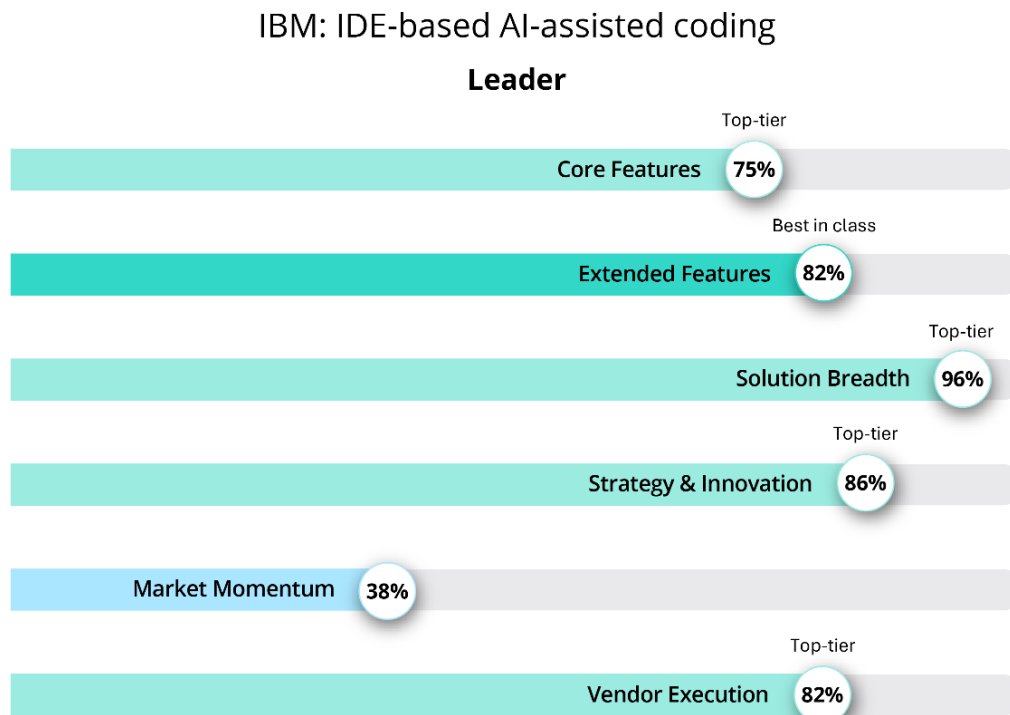
It supports in-place modernization and optimization, leveraging existing programming languages or transforming to Java-based, well-architected, oriented solutions. Hybrid, on-premises, and air-gapped deployment options for regulated industries are expected in 3Q26.

Bob also supports agentic experiences that converge understanding, explanation, generation, refactoring, transformation, and testing into unified workflows, going beyond generic assistants by handling complex, context-heavy enterprise scenarios like monolith-to-microservices conversions, framework migrations (e.g., Java upgrades, Struts or JSF to React or Angular), and full SDLC automation. While still emerging publicly (with expansions planned for broader language support, deeper integrations, and more autonomous agents), Bob excels in regulated, large-scale environments where trust, governance, and enterprise expertise are critical. Overall, IBM Bob stands out in the following:

- Enterprise application modernization
- Security
- IP protection

- Deep agentic automation
- Broad insight through static
- Dynamic code analytics capabilities
- Full-lifecycle coverage in regulated and large-scale environments

Figure 8: Omdia Universe ratings—IBM



Source: Omdia

Strategy & Innovation **Strengths**

- It strongly emphasizes responsible AI and multi-model orchestration, optimizing for accuracy, performance, and cost.
- It offers fine-tuning for accuracy and hallucination reduction.
- It excels in agentic AI workflows, especially for mainframe and legacy codebases.
- It offers core differentiation in specialized modernization, such as COBOL to Java.
- It provides static and dynamic code insights to provide enterprise-wide visibility.

Market Momentum

Strengths

- A long-established vendor with massive global scale (thousands of partners and SIs, coverage in all major regions, it has strong partnerships like AWS, Microsoft, and Red Hat (which it owns).
- It targets large enterprise accounts with direct sales and tech partnerships.
- It is a leader in relevant analyst views.

Limitations

- Bob was launched GA at the time of this report's publication, but IBM has not disclosed customer accounts at this time.

Vendor Execution

Strengths

- It offers excellent enterprise-grade execution with robust security capabilities.
- It features automated LLM selection for cost benefit.
- It is in high availability and scaling via IBM Cloud.
- It provides enterprise modernization at scale.
- It also offers hybrid, on-premises, and air-gapped options for deployment.

Limitations

- Billing and SLA resolution time details are currently limited.
- Peer ratings are also currently limited.

Solution Breadth

Strengths

- Language support (including rare legacy languages like COBOL, RPG, PL/I) is extremely wide.
- It covers the full SDLC, strong in planning, design, coding, testing, documentation, maintenance, and modernization.
- It offers strong context awareness via RAG codebase ingestion.
- It provides NLI and role-based access.

Limitations

- Analytics, dashboards, and performance monitoring are more modernization-focused than comprehensive end-to-end KPIs.
- It has a catalog of components and connectors, but it is not extensive.

Core Features

Strengths

- Coding, including context from the IDE and the codebase, is outstanding.
- It creates snippets to large-scale generation, refactoring, and code optimization.
- It emphasizes secure, enterprise-grade code automation.
- It excels in testing through NL and unit, integration, security testing, and edge test cases, and generates executable tests.
- Documentation, with inline comments, READMEs, and examples, is strong.
- It offers legacy language support.
- It performs code reviews for performance, refactoring, and security.

Limitations

- It cannot identify bottlenecks such as single points of failure and scope churn.

Extended Features

Strengths

- API and data services such as REST, SOAP, GraphQL, gRPC, OpenAPI, auto SQL, and mock APIs are very strong.
- It has a broad database and cloud support.
- Security, privacy, and compliance, including GDPR, PCI, HIPAA, SSO, SAML, MFA, vulnerability scanning, and confidential computing, are excellent.
- AI capabilities, such as code transformation, modernization, guardrails, and hallucination minimization, are powerful.

Appendix

Methodology

Omdia Universe

Omdia's rigorous methodology for a Universe involves the following steps:

- Omdia analysts perform an in-depth review of the market using Omdia's market forecasting data and Omdia's enterprise insights survey data.
- Omdia creates a matrix of capabilities, attributes, and features that it considers to be important now and in the next 12–18 months for the market.
- Vendors are interviewed and provide in-depth briefings on the current solutions and future plans.

- Analysts supplement these briefings with other information obtained from industry events and user conferences.
- The Universe is peer-reviewed by other Omdia analysts before being proofread by a team of dedicated editors.

Inclusion criteria

Vendors are invited to participate in this Universe if they have a recognized solution in the market that is widely available geographically and suitable across industries. Startups are included if they have exceptionally strong solutions and have the potential to become market leaders.

Universe assessment

Omdia provides a high-level summary. Below are topics that reflect key solution capabilities and that drive Omdia's assessment of the AI-assisted software development solution comparison.

Solution capability

The section above titled "Analyzing the AI-assisted Software Development: IDE-based Tools Universe" provides a summary of the solution capabilities assessed on the x-axis of the Universe chart.

Strategy and execution

The aspects assessed on the y-axis of the Universe chart are given below.

Market momentum

- Company and product history and maturity
- Net Promoter Score (NPS), if available
- Global reach
- Vertical and horizontal focus
- Revenue and annual growth
- High-level customer-related questions

Strategy & innovation

- Technological innovation
- Roadmap
- Licensing
- Target market and users
- Deployment options
- Competitive differentiation

Vendor execution

- Partner and systems integrator ecosystem
- Technology partners
- Customer support
- ROI
- Peer-to-peer customer score

Market penetration

The size of the circle (z-axis) on the Universe chart represents the following metric:

- Revenue-based metric

Further reading

[*Omdia Universe: No-Low-Pro AppDev Platforms, 2025*](#) (May 2025)

[*Omdia Universe: No-Low-Pro IDE Assistants, 2025*](#) (May 2025)

[*Omdia Market Landscape: AI-Assisted Software Testing*](#) (September 2025)

[*Omdia Market Radar: AI-Assisted Software Development, 2023–24*](#) (January 2024)

[*Omdia Universe: No Code, Low Code Solutions, 2023–24*](#) (September 2023)

[*Generative AI: The impact of AI-based autocoding on software development*](#) (March 2023)

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